Emotion-Driven Common Sense Retrieval for More Natural Language Processing

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In our research we concentrate on Common Sense which is often called "The Holy Grail of Artificial Intelligence" although we strongly believe that emotions are equally important for machines to understand human's behavior. This paper introduces our thoughts about mutual struggle between common sense and emotions, their influence on each other and combines these thoughts with Bentham's "felicific calculus" which suggested us new ideas to continue exploring fields of common sense and affective computing. We will underline the need of machines understanding the basic human feelings which we presume to be the base of our every behavior and suggest how this goal could be technically achieved. We will also try to convince the readers and workshop participants that machines equipped with folk logic (or folk beliefs, folk knowledge) might be much more natural (also in the aspect of language usage) than systems using traditional, closed and strict resources and methods often developed by great thinkers.

Keywords: Affective Computing, Commonsense Processing, Web-mining

1 INTRODUCTION

1.1 Need for Naturally Talking Machines

The machines around us are getting more and more sophisticated but we do not always use their full potential while a cellural phone manual can have hundreds of pages and not many people have time and will to learn it all. It would be much easier if we could explain our device what we want from it. Therefore the easiest way to do it is to explain everything in natural language which we use when we talk to other people. However, while setting up things which are important for us, we would rather not trust the machines which do not understand basic things about us, human beings, and our everyday life. If we let's say at 11PM, accidentally set an alarm clock to 9PM instead 9AM, we wish the machine discovered our mistake before we oversleep. Knowing each other leads people to trust and the most natural way of knowing another person is communicating with that person in a plain, natural language. Achieving such naturalness is a goal for many researchers already, but we think they take a roundabout way. Current interfaces for human-computer communication are being developed to *behave* as naturally as possible, their eyes follow our faces, artificial hands try to gesticulate, faces expressions try to react when hearing our laughter [1][2]. However, they still lack what is most important for natural communication – the naturalness of their language responses in the semantical level when the topic of the talk is open. We claim that without achieving commonsensical knowledge and understanding our emotional mechanisms it will be very difficult for a machine to reason properly about human beings.

1.2 Our Approach to AI

The most popular objection made by the adversaries of Artificial Intelligence and systems behaving like humans is that a machine will never be able to go beyond the borders created by a programmer, that computers will always be only the prisoners of somebody's algorithm and personally chosen data. However our approach is to build a system where not a programmer but real world information will decide most of its behaviors. Therefore we develop methods using natural language and very large unannotated corpora (the Web in particular) to let the machine acquire commonsensical knowledge needed in given situation.However, still the output is too poor to develop a naturally reasoning conversation system. The Internet is certainly very convenient text resource for inductive or abductive learning but we felt that many problems with common sense could be fought over by implementing our methods with emotion-based mechanisms letting a machine fill the gaps in its common sense reasoning and predicting user behaviors and motivations.

2 DEFINITIONS WE USE

2.1 What is Common Sense?

The term "common sense" is, in the broadest meaning, what people in common would agree, what they believe. Most people's experience which does not depend upon esoteric knowledge or study or research, but is based upon what is believed to be knowledge held by people "in common" (folk knowledge. folk logic). The knowledge and experience most people have, or are believed to have by the other person, though changing from culture to culture and environment to environment. We claim that a machine should have at least the set of commonsensical knowledge needed

for a given situation and acquire new or update old set when the situation changes.

2.2 What are Emotions?

Because "emotions" may be defined as a neural impulse that moves an organism to action prompting automatic reactive behavior that has been adapted through evolution as a survival mechanism to meet a survival need, we rather concentrate on feelings as the inside states which are expressed outside through physiological functions such as facial expressions, faster heartbeat, and behaviors such as aggression or crying. We simplify the feelings to "pleasant" and "unpleasant" (pain), however we also presume that one day machines should be emotionally intelligent as Mayer et al.[3] suggested, and achieve:

- The capacity to accurately perceive emotions.
- The capacity to use emotions to facilitate thinking.
- The capacity to understand emotional meanings.
- The capacity to manage emotions.

We put stress on the perceiving and understanding emotional information - using and managing functions will be needed later when constructing human-computer interfaces.

2.3 Emotions vs. Common Sense

We believe that human inborn affective mechanisms (instincts, physical reactions, etc.) are constantly influenced by and influence the common sense knowledge. Lately, the latter was proved empirically[4]. Economists and neuroscientists at two American universities, Carnegie Mellon University and the University of Iowa performed experiments to see whether people's feelings got in the way of their intellects. They discovered that stroke victims and people with damaged emotional centers of their brains made better investment decisions than those with a full emotional capabilities. The researchers gave the subjects $20 and asked them to invest – that is, bet on the flip of a coin, $1 at a time. Heads, they lost $1; tails, they ended up with $2.50. On any flip, they had the right for declining to invest and simply keep the dollar. Given the fifty-fifty odds of a coin toss, a subject would earn $25 by gambling every time but end up with only $20 for consistently keeping the money. Healthy subjects tended to hold on their dollars, investing only 58% of the time. Subjects with brain damage bet on 84% of the flips and earned more money. This is why we claim that all our behaviors depend on the battle between emotions and common sense.

3 METHODS WE USE

3.1 How to Predict Human Behaviors

There are many researchers who performed or perform their works on common sense, often called "The Holy Grail of Artificial Intelligence". Many more and less famous ideas and concepts were developed but for predicting previous and following actions, their starting points and goals, we concentrated on Schankian scripts. In his classic book [7], followed by [8] Schank has introduced inquiry into the nature of knowledge that is needed to understand the world and understand natural language. His main claim is that structured knowledge dominates understanding while the question is the content of these structures. He takes a pragmatic approach that does not separate form from content. He proposes conceptual primitives we decided to partially adopt, although his proposal has not been tested psychologically. Schank concentrates on memory, and in particular memory organization but we use only our original Bacterium Lingualis Memory concept[10], which will not be mentioned here. He argues, and we agree with him, that understanding language involves causally connecting thoughts/sentences. Because causality is often implied or incompletely described, it is usually harder to understand connected text than individual sentences. Therefore, he describes a formal representation (causal syntax) of causal chains which is to the discourse level what Conceptual Dependency is to the sentence/thought level. Rules are provided in which every primitive action is associated with the set of states it can affect as well as those that enable it. Schank describes scripts as groups of causal chains that represent knowledge about frequently experienced events (the most famous example is "going to a restaurant"). In other words, a script is a stereotyped sequence of actions that defines a well-known situation and has associated with it: a number of roles for the actors (different points of view on the situation, e.g. customer vs waiter vs cook), different tracks (e.g. restaurant, fast-food), different scenes (e.g. enter, order, eat, pay); each scene has a main conceptualization, which must have happened if the scene is instantiated, as well as props, entry conditions, results, branches and loops etc.

3.2 Automatic Retrieval of Schankian Scripts

This part is mainly based on the thoughts of Wilensky [9] who was influenced by Schank but introduced goals and planning in a deeper manner. Schank and Wilensky have suggested that a need makes plan to achieve a goal and gave us wide range of propositions how the machine could use the simplest and the shortest ways to reach the target. However such suggestions fit the mobile robots research not the robots which try to understand human commonsensical behavior. We claim that instincts and feelings are the elements that influences our behaviors and the way in which we change our goals, plans what modifies the scripts. The closest one is restaurant A but one may choose restaurant B because its staff works faster. There are many probably elements which could have stronger influence on our choices than hunger. Our task is to predict as many possibilities as possible. Scripts are important

part of our basic thinking processes and can be triggered as associations to any word we are told. This supposed that this is the first step for creating machine imagination. We also perform experiments on causal dependencies (IF-forms), which are retrieved in almost identical way as scripts. Such a system's architecture can be summarized into the following processing steps:
- A noun is assigned for a **keyword** (any keyword spotting method can be applied);
- The system uses our web corpus for frequency check to retrieve **3 most frequent verbs** following the keyword noun;
- The **most frequent particle** between noun keyword and 3 most frequent verbs is discovered;
- For creating bi-gram the system retrieves a list of most frequent **verbs** occurring **after** the previously chosen **verb**;
- By using Google search engine[5], the system checks if the noun-particle unit occurs with new verb-verb unit for **time-sequence actions** and verb-if unit for **casual dependencies**;
- If yes - the VO-then-V and VO-if-VO **units are stored:**

$$\mathbf{VO_{then}V} = \mathbf{N} + \mathbf{P_{max}} + \mathbf{V_{max1}} + \mathbf{V_{max2}}$$

N:Triggering noun (keyword);
P_{max}*:most frequent particle joining noun and verb;*
V_{max1}*:most frequent verb occurring after the N;*
V_{max2}*:most frequent verb occurring after* V_{max1}*;*

$$\mathbf{VO_{if}V} = \mathbf{N_1} + \mathbf{P_{1max}} + \mathbf{V_{1max}} + if + \mathbf{N_2} + \mathbf{P_{2max}} + \mathbf{V_{2max}}$$

N_1*:Triggering noun (keyword);*
P_1max*:most freq. particle joining first noun with a verb;*
V_1max*:most freq. verb after the* N_1P_1*;*
N_2max*:most freq. noun after* $N_1P_{1max}V_{1max}$ *and "if";*
P_2max*:most freq. particle joining* N_2 *and* V_2*;*
V_2max*:most freq. verb after* N_2P_2*;*

To retrieve minimal scripts (mentioned in the Theory Section) consisting of at least two actions following each other, we created a program which performs following steps:

- Search the most frequent particle following noun keyword in Noun-Particle Dictionary;
- Search three most frequent verbs from Noun-Particle-Verb dictionary as the first actions using Noun keyword and the most frequent particle;
- if the Verb of Noun-Particle-Verb is verb keyword, the script is a minimal script; if not, search the most three nouns using the verbs from Noun-Particle-Verb dictionary and search the most three verbs from Verb-Verb Dictionary as the second action using the first verbs.
  if the verb is not followed by any actions, the script is be deleted;
- repeat previous step until the last action is verb keyword.

4 WHAT WE PROPOSE

Although we confirmed that above mentioned algorithm works and is able to discover valuable knowledge[12] it cannot be described as flexible, as longer scripts need learning, categorization, similarity calculations and lose their certainty degree as being estimated instead of retrieved. Then we took part in the discussion about machine ethics[11] which gave us an idea to refresh and reuse the concept of Positiveness[?] which could lead us to creating an affective mechanism which coexists with everyday knowledge we use. To mixing emotions with common sense as mutual partners controlling each other. Five level Positiveness degree was able to guess the emotional load of words and very short expressions but we need more complex affective processing for supporting Schankian scripts retrieval and analysis. We decided to use over two hundred years old ideas of Jeremy Bentham, one of the fathers of utilitarianism. In his book[6] he wrote: "Nature has placed mankind under the governance of two sovereign masters, pain, and pleasure. It is for them alone to point out what we ought to do, as well as to determine what we shall do." He understood "pleasure" broadly including all pleasant feeling or experiences[1] while "pain" includes all unpleasant feeling or experiences[2]. We presume that our lives is based on pain and pleasure and we are under their control even without expressing it in natural language. Pleasure and pain themselves might be states, states of affairs, things, events or properties. Whatever a human does is motivated for keeping a balance between these two. Bentham formulated an algorithm called "felicific calculus" for calculating the degree or amount of happiness that a specific action is likely to cause, and hence its degree of moral rightness. Felicific calculus is supposed to establish the moral status of any considered act. Variables were set as follows:

- intensity
- duration
- certainty or uncertainty
- propinquity or remoteness
- fecundity – the probability it has of being followed by sensations of the same kind
- purity: the probability it has of not being followed by sensations of the opposite kind
- extent: the number of persons to whom it extends

[1] Such as elation, ecstasy, delight, joy, and enjoyment

[2] Such as aches, throbs, irritations, anxiety, anguish, chagrin, discomfort, despair, grief, depression, guilt and remorse.

As Bentham's proposal was not easy to remember he proposed a "mnemonic doggerell":

Intense, long, certain, speedy, fruitful, pure?
Such marks in pleasures and in pains endure.
Such pleasures seek if private be thy end:
If it be public, wide let them extend
Such pains avoid, whichever be thy view:
If pains must come, let them extend to few.

Our idea is to calculate actions which make Schankian scripts using Bentham's calculus in order to fill gaps in machine reasoning. Any robot, even having a full script which human uses to perform some tasks should know what are the deeper goals and motivations of humans. Not only "what do we do at the restaurant" but also "why do we go to restaurants" or "why do we prefer fancy restaurant". We think that such emotion understanding is a key to more natural machines.

5 CONCLUSIONS

There are two main reasons for which the authors decided to introduce a research before beginning its development. First one is to spark a discussion about our affective approach to common sense among national researchers. Because of the lack of affective computing specialists around us, the Workshop is a great opportunity to hear and meet specialists not only from NLP field but also cognitive science, linguistics and maybe philosophy, psychology or sociology fields. This is the second reason. We performed several experiments with commonsensical knowledge and we felt the lack of goal understanding while the program was analyzing human's everyday behaviors. We suggest defining the emotion mechanism based on Bentham's felicific calculus which could describe all human by default. Even if ethical theories are often controversial we would like to propose a new trend in common sense processing which can be helpful in learning and retrieving everyday knowledge.

6 FUTURE WORK

We plan to build a prototype which will use Bentham's calculus and hope to present the results of preliminary experiments during our presentation. We would like to prepare a modern version of 18th century philosopher's algorithm and prove that our assumptions made in this paper are correct and useful for the NLP community. We will try to show that "folk knowledge" with an basic "positive / negative" ethical engine performs very natural reasoning.

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